

Dr. Aayushi Tandon

Climate Scientist | AI/ML for Weather & Climate |

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https://scholar.google.co.uk/citations?user=9u_DDsAAAAAJ&hl=en

PROFESSIONAL SUMMARY

Climate scientist with a PhD in Climate Science and a sustained research focus on applying machine learning to extreme precipitation, multi-model ensemble evaluation, and hydro-meteorological hazard prediction across the Indian subcontinent. Builds end-to-end AI/ML pipelines - deep learning, ensemble methods, statistical downscaling and bias correction - for terabyte-scale CMIP6 and NWP ensemble datasets in GPU accelerated environments. Track record of nine peer-reviewed publications, conference recognition, graduate student mentoring, and contributions to national and international research funding proposals. Member of AGU, EGU and AOGS, with an established international research network. Seeking to bring this technical depth and growing research-leadership experience to a mission-driven team advancing operational AI weather and climate forecasting.

CORE EXPERTISE

- **AI/ML for Weather & Climate:** deep learning (CNNs, encoder-decoder ANNs), ensemble methods (Random Forest, Gradient Boosting), statistical downscaling, bias correction of NWP and operational ensemble forecasts
- **Climate & Hydrological Data:** CMIP6 multi-model ensembles, extreme precipitation and flood-risk analysis, process-based hydrological modelling (HEC-HMS, HEC-RAS)
- **Computing & Tools:** Python (advanced), R, MATLAB; TensorFlow, Keras, PyTorch, Scikit-learn; large-scale GPU accelerated pipeline design; ArcGIS, QGIS, Google Earth Engine
- **Research Leadership:** grant proposal contribution, graduate student supervision, cross-disciplinary and international collaboration, science communication to policy and non-specialist audiences

RESEARCH EXPERIENCE

Post-Doctoral Scientist - National Institute of Technology Delhi

May 2026 - Present

- Leading research applying deep learning and statistical ML frameworks to downscale and post-process CMIP6 ensemble outputs into decision-relevant regional climate projections for India, extending doctoral research into operational climate services.
- Architecting end-to-end Python workflows for terabyte-scale climate data processing and ML-driven extreme-event attribution within HPC environments.
- Co-supervising graduate students on climate-data-science projects while carrying weekly teaching load across B.Tech/M.Tech programmes.
- Contributing to national and international grant proposals and funding applications in partnership with senior faculty.

Project Scientist, ANRF-PAIR - Berhampur University, Odisha

Jan 2026 - Apr 2026

- Designed and implemented AI/ML models (Scikit-learn, TensorFlow) for air-pollution prediction and exposure mapping across Indian megacities, applying precipitation-postprocessing methods directly transferable to ensemble forecast analysis.
- Processed large-scale environmental datasets with GPU-scale pipelines and ensemble model outputs to identify national-level pollution patterns.
- Built data-visualisation dashboards translating complex model outputs into actionable insight for government and policy stakeholders.

Doctoral Research Scholar - University of Petroleum and Energy Studies (UPES), Dehradun 2022 - 2026

Thesis: "Fidelity of CMIP6 Models using Machine Learning in Projecting Extreme Precipitation Events Over North Indian States"

- Evaluated CMIP6 multi-model ensemble fidelity for extreme precipitation over North India, establishing a regional benchmark for model performance in monsoon-dominated environments - directly relevant to operational forecast verification.
- Developed and benchmarked ML algorithms (Random Forest, Neural Networks, Gradient Boosting) for precipitation pattern attribution and projection under SSP3 and SSP5 emission scenarios, achieving 85%+ predictive accuracy.
- Engineered end-to-end Python pipelines for terabyte-scale CMIP6 processing in GPU accelerated environments, reducing computation time by 40% - demonstrating operational readiness for large-scale ensemble postprocessing.

SELECTED PUBLICATIONS

- Tandon, A., Awasthi, A., & Pattnayak, K. C. (2026). Integrating IMDAA Regional Reanalysis and Machine Learning for Enhanced Detection of Extreme Precipitation Over Complex Himalayan Terrain. *Earth Systems and Environment*.
- Tandon, A., Kishor, K., Awasthi, A., & Pattnayak, K. C. (2026). Assessment of precipitation extremes and risk factors in the Himalayan foothills: a machine learning approach for hydro-meteorological hazard analysis. *Natural Hazards*, 122(3), 92.
- Tandon, A., Awasthi, A., & Pattnayak, K. C. (2025). Extreme Precipitation Projections in Asian Subregions: Study on CMIP6 Development, Geographic Patterns, and Research Gaps. *Physics and Chemistry of the Earth, Parts A/B/C*.
- Tandon, A., Awasthi, A., & Pattnayak, K. C. (2025). Efficacy of machine learning in simulating precipitation and its extremes over the capital cities in North Indian states. *Scientific Reports*, 15(1), 10345.
- Awasthi, A., Pattnayak, K. C., Tandon, A., Sarkar, A., & Chakraborty, M. (2023). Implications of climate change on surface temperature in North Indian states: evidence from CMIP6 model ensembles. *Frontiers in Environmental Science*, 11.

Enclosed publication details

LEADERSHIP, GRANTS & COLLABORATION

- Contributed to the development of national and international research funding proposals alongside senior faculty across two institutions.
- Co-supervised and mentored graduate students on applied climate-data-science projects; delivered structured university teaching (8 credits/week).
- Built cross-institutional research collaborations spanning UPES, NIT Delhi, Berhampur University and IITM Pune; presented findings at AGU, EGU-affiliated and national forums.
- Active member of the American Geophysical Union (AGU), European Geosciences Union (EGU), and Asia Oceania Geosciences Society (AOGS).

EDUCATION

Ph.D. in Climate Science - University of Petroleum and Energy Studies (UPES), Dehradun, 2022-2026

M.Sc. in Physics - HNB Garhwal University, Srinagar, 2020-2022 (CGPA: 6.96, First Class)

B.Sc. (Hons) in Physics - Graphic Era Hill University, Dehradun, 2017-2020 (CGPA: 9.11, First Class with Distinction)

AWARDS & RECOGNITION

- Best Presentation Award - International Conference on Machine Learning in Sciences (ICMS), UPES, 2025
- Best Presentation Award - DREAMS International Conference, 2025
- Student Travel Grant - American Geophysical Union (AGU) Annual Meeting, Washington DC, 2024
- Speaker Felicitation - Annual Monsoon Workshop and National Heatwave Symposium, IITM Pune, 2024

SELECTED PROFESSIONAL DEVELOPMENT

- ICTP Summer School: Theory, Mechanisms and Hierarchical Modelling of Climate Dynamics - Artificial Intelligence and Climate Modelling, 2025
- Hydrological Modelling: HEC-HMS, HEC-RAS and Machine Learning (HyMHECMaL 2025) - National Institute of Hydrology, Roorkee
- ICTP Workshop on Global Monsoons: Theory, Models and Observations, 2025
- Certificate Course: Satellite Data Products for Climate Change Applications, 2025

PUBLICATIONS

Published Manuscripts

1. Tandon, A., Awasthi, A., & Pattnayak, K. C. (2026). Integrating IMDAA Regional Reanalysis and Machine Learning for Enhanced Detection of Extreme Precipitation Over Complex Himalayan Terrain. *Earth Systems and Environment*, 1-26. (10.1007/s41748-026-01117-3)
2. Tandon, A., Kishor, K., Awasthi, A., & Pattnayak, K. C. (2026). Assessment of precipitation extremes and risk factors in the Himalayan foothills: a machine learning approach for hydro-meteorological hazard analysis. *Natural Hazards*, 122(3), 92.
3. Tandon, A., Awasthi, A., & Pattnayak, K. C. (2025). Extreme Precipitation Projections in Asian Subregions Regions: Study on CMIP6 Development, Geographic Patterns, and Research Gaps. *Physics and Chemistry of the Earth, Parts A/B/C*, 104148.
4. Tandon, A., Awasthi, A., Pattnayak, K. C., & Das, S. (2025). Rainfall variability and rising extremes in urbanizing himalayan foothills: a machine learning and data-driven exploration of hydroclimatic shifts in Uttarakhand, India. *Earth Systems and Environment*, 1-26.
5. Tandon, A., Awasthi, A., & Pattnayak, K. C. (2025). Efficacy of machine learning in simulating precipitation and its extremes over the capital cities in North Indian states. *Scientific Reports*, 15(1), 10345.
6. Tandon, A., Awasthi, A., Pattnayak, K. C., Tandon, A., Choudhury, T., & Kotecha, K. (2025). Machine learning-driven solar irradiance prediction: advancing renewable energy in Rajasthan. *Discover Applied Sciences*, 7(2), 107.
7. Awasthi, A., Pattnayak, K. C., Tandon, A., Sarkar, A., & Chakraborty, M. (2023). Implications of climate change on surface temperature in North Indian states: evidence from CMIP6 model ensembles. *Frontiers in Environmental Science*, 11, 1264757.
8. Tandon, A., Awasthi, A., & Pattnayak, K. C. (2023, November). Comparison of different machine learning methods on precipitation dataset for Uttarakhand. In *2023 2nd International Conference on Ambient Intelligence in Health Care (ICAIHC)* (pp. 1-6). IEEE.

Communicated and Under Review Manuscripts

9. Tandon, A., Awasthi, A., Sahu, S., Pattnayak, K. C. & Das, S. (2026). Non-linear Intensification of Extreme Precipitation and Concurrent Drought Risk in North India Under 1.5°C - 4.0°C Warming: A Machine Learning Projection Framework
10. Tandon, A., Sahu, S. (2026). Meteorological Controls on PM2.5 and Associated COPD Burden in Delhi, India: A Spatiotemporal Analysis Using Bidirectional LSTM with Attention Mechanism and SHAP Interpretability
11. Tandon, A., Kumar, P., Awasthi, A., & Pattnayak, K. C. (2026). Identifying Best-Performing CMIP6 Models for Extreme Precipitation Simulation over the Indo-Gangetic Plain using Multi-Metric Skill Assessment